

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2024 Course: B.Tech Branch: Computer Engineering/Computer Science & Engineering Semester : VI Subject Code & Name: BTCOC601_Y23 Compiler Design Max Marks: 60 Date: 18/12/2024 Duration: 3 Hr.			
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	A compiler for a high-level language that runs on one machine and produces object code for a different machine is called a. Multipass Compiler b. Cross Compiler c. One pass Compiler d. Two-pass Compiler		1
2	An optimizing compiler, a. is optimized to take less execution time. b. is optimized to take less space. c. optimizes code. d. none of the above.		1
3	What does a compiler do upon encountering an error during compilation? a. Produce warnings. b. Terminate compilation. c. Complete reading the program. d. Terminate program.		1
4	Which among the following is an abstract form of intermediate code? a. Object code. b. Byte code. c. 3-address code. d. None of the above		1
5	What is the function of intermediate code? a. To make errors easily detectable. b. To eliminate the need for a new full compiler for every unique machine. c. Do semantic analysis. d. None of the above.		1
6	Among the parsing techniques, which avoids backtracking? a. Recursive-descent parsing & predictive parsing. b. Top-down parsing. c. Predictive parsing. d. Recursive-descent parsing		1
7	Which symbol table implementation is based on the property locality of reference?		1

	a. Search Tree. c. Hash table.	b. Self-organizing list. d. Linked list.		
8	In which phase in compiler design do we check the grammar of the programming language? a. Lexical analysis. c. Syntax analysis.	b. Semantic analysis. d. Code generation.		1
9	A top-down parser generates a. Left-most derivation in reverse c. Right-most derivation in reverse derivation	b. Left-most derivation d. Right-most derivation		1
10	Which table is a permanent database that has an entry for each terminal symbol? a. Reductions c. Literal table	b. Identifier table d. Terminal table		1
11	The most general phase structured grammar a. Context sensitive c. Regular	b. Context free d. None of these		1
12	The phase Syntax Analysis is modeled on the basis of a. High level language c. Context free grammar	b. Low level language d. Regular grammar		1
Q.2	Solve the following.			12
A)	Explain the major activities performed during the lexical, syntactic, and semantic analysis phases. How source program is analyzed?	Understand Remember		6
B)	How many phases are there in a compiler? Explain each phase in detail?	Remember Understand		6
Q.3	Solve the following.			12
A)	Explain syntax directed definition with simple examples? State the applications of Syntax Directed Definition.	Remember Knowledge		6
B)	Write the properties of LR parser with its structure. Also explain the techniques of LR parser.	Understand Apply		6
Q.4	Solve Any Two of the following.			12
A)	Define LL(1) also Differences between SLR, CLR, LALR parsers? State the problems in Top Down Parsing?	Remember Knowledge		6
B)	Write the definition of symbol table and procedure to store the names in symbol table. What are the data structures used in symbol table?	Understand Apply		6
C)	What is bottom-up parsing? Describe the main techniques used in bottom-up parsing, including shift-reduce parsing.	Remember Knowledge		6
Q.5	Solve Any Two of the following.			12

A)	What is intermediate code? Explain within example and write the benefits of intermediate code generation.	Remember Understand	6
B)	Consider the following grammar: E → E+T T T → T*F F F → F^1/b Construct the SLR parsing table and also parse the input "a*b+a".	Knowledge Apply	6
C)	How a graph representation of an intermediate code is done? Explain the concept of basic blocks and flow graphs. Also, convert an example source code to the flow graph.	Remember Apply	6
Q. 6 Solve Any Two of the following.			12
A)	What are the properties of code generation phase? Also explain the Design Issues of this phase.	Remember Understand	6
B)	What do you mean by code optimization? Explain the concept of machine dependent and independent code optimization.	Remember Understand	6
C)	Explain the concept of DAG representation of basic blocks. What is the optimization of basic blocks?	Knowledge Understand	6
*** End ***			